

Properties of logarithms



1 Simplify each of the following expressions:

a $\log 4 + \log 9$

c $\log_{10} 11 + \log_{10} 2 + \log_{10} 9$

e $\frac{\log_{10} 4}{\log_{10} 2}$

g $\frac{\log a^8}{\log a^4}$

i $\log 10x + \log 10y$

k $\frac{\log \left(\frac{1}{x^4} \right)}{\log x}$

m $\frac{5 \log m^2}{6 \log \sqrt[3]{m}}$

o $10^{\log w}$

b $\log_{10} (10) + \log_{10} (10)$

d $\log_{10} 12 - (\log_{10} 2 + \log_{10} 3)$

f $\frac{\log_4 125}{\log_4 5}$

h $\frac{\log a^3}{\log \sqrt[3]{a}}$

j $\log x^2 + \log x^3$

l $\log_{10} 10 + \frac{\log_{10} (15^{20})}{\log_{10} (15^5)}$

n $\frac{8 \log_{10} (\sqrt{10})}{\log_{10} (100)}$

p $x^{4 \log_x 3 - 6 \log_x 2}$

2 Simplify each of the following expressions:

a $\frac{\ln a^6}{\ln a^3}$

b $\frac{12 \ln (m^4)}{10 \ln \sqrt{m}}$

c $\frac{\ln (a^4)}{\ln \sqrt[3]{a}}$

d $\frac{2 \ln x}{10 \ln y}$

3 Rewrite the following as the sum or difference of logarithms without any powers, fractions or radicals:

a $\log_9 uv$

b $\log (x^4)$

c $\log_4 (x^7)$

d $\log_5 \left(\frac{9}{7} \right)$

e $\log \left(x^{\frac{2}{5}} \right)$

f $\log_b (x^2)$

g $\log (3x^{-1})$

h $\log \left((3x)^5 \right)$

i $\log (7x^{-4})$

j $\log \left((5x)^{-7} \right)$

k $\log \left((2x)^{-1} \right)$

l $\log \left(\frac{1}{xy} \right)$

m $\log \left(\frac{pq}{r} \right)$

n $\log \left((3x + 7)^{-1} \right)$

o $\log \left((3x + 4)^{-8} \right)$

p $\log \left((5x + 7)^{\frac{1}{2}} \right)$

q $\log \left((5x + 2)^6 \right)$

r $\log \left((x + 6)^5 \right)$

s $\log_2 (5x)$

t $\log_{10} \left(\frac{2}{x} \right)$

u $\log (m^2)$

v $\log \left(5x^{\frac{2}{3}} \right)$

w $\log \left((14x)^{\frac{1}{3}} \right)$

x $\log \left(\sqrt{\frac{c^8}{d}} \right)$



- 4 Rewrite the following as the sum or difference of logarithms without any powers, fractions or radicals:

a $\ln\left(\frac{15}{8}\right)$

b $\ln\left(\frac{pq}{r}\right)$

c $\ln\left(\frac{19}{7}\right)$

- 5 Rewrite each of the following as a single logarithm or integer:

a $5 \log x^3 - 4 \log x^2$

b $5 \log x + 3 \log y$

c $8 \log x - \frac{1}{3} \log y$

d $7 \log x - \log\left(\frac{1}{x}\right) - \log y$

e $7 \log_{10} 5 - 21 \log_{10} 25$

f $5 \log_{10} 8 - 3 \log_{10} 4$

g $2 \log_6 3 + \frac{1}{3} \log_6 64$

h $\log_2 36 - 2 \log_2 3$

- 6 Rewrite each of the following expressions as a single logarithm:

a $\ln 3 + \ln 5$

b $\ln 24 - \ln 4$

c $\ln 5 - \ln 7$

d $\ln 11 + \ln 5 + \ln 8$

e $\ln 24 - (\ln 2 + \ln 6)$

f $\ln 7 + \ln 2 + \ln 3 - \ln 5$

- 7 Rewrite each of the following expressions as a single logarithm:

a $6 \ln x - \frac{1}{6} \ln y$

b $\ln(3x) + \ln(5y)$

c $\ln(x^2) + \ln(x^3)$

d $3 \ln(x^5) - 4 \ln(x^2)$

- 8 Use the properties of logarithms to express each of the following without any powers or radicals:

a $\ln y^2$

b $\ln \frac{1}{y^2}$

c $\ln \sqrt{y}$

d $\ln \sqrt[3]{y}$

e $\ln y^{-5}$

f $\ln y^{\frac{2}{9}}$

g $\ln y^{\frac{1}{2}}$

h $\ln \sqrt{w}$

- 9 Use the properties of logarithms to rewrite each of the following in the form $\ln b^a$ where $a > 0$:

a $-2 \ln x$

b $\frac{\ln x}{2}$

Change of base



10 Rewrite the following in terms of base 10 logarithms:

a $\log_4 16$

b $\log_3 0.9$

c $\log_3 \sqrt{5}$

d $\log_a B$

11 Rewrite $\log_3 20$ in terms of base 4 logarithms.

12 Consider the following logarithmic expressions:

i Rewrite the expression in terms of base 10 logarithms.

ii Now, evaluate each to two decimal places.

a $\log_8 21$

b $\log_4 \sqrt{5}$

c $\log_2 3^3$

d $\log_\pi 105$